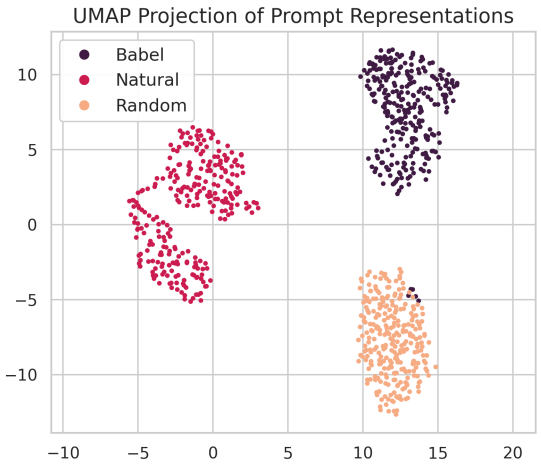

Figure 4. U-map of the last hidden state representations of LLaMA2-7B for successful Babel, natural and random prompts constructed for all four datasets.

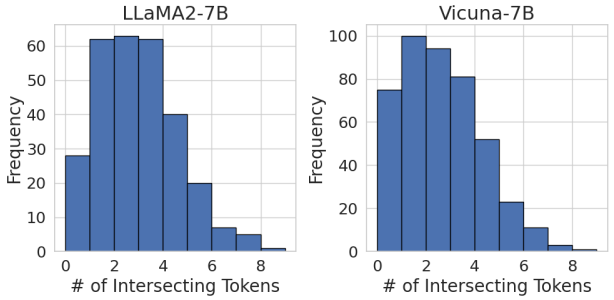

Figure 5. Histogram of the number of shared tokens in the target text and Babel prompts.

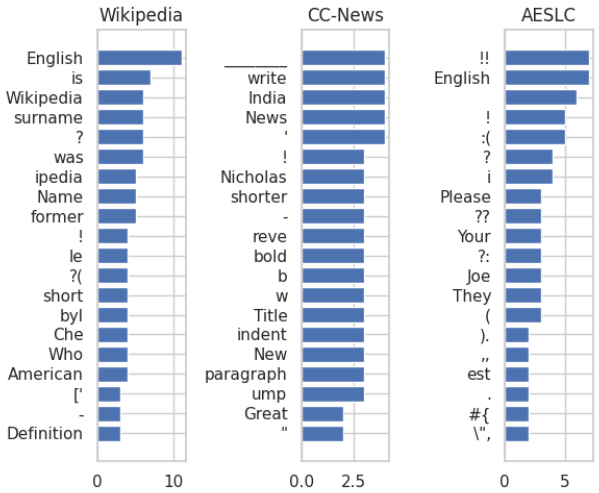

Figure 6. The most frequent tokens in Babel prompts optimized for Vicuna-7B model.

influenced by the model's self-attention mechanisms. The transformed representations, visualized using Uniform Manifold Approximation and Projection (UMAP) in Figure 4 and Appendix Figure 8, reveal distinct clustering patterns for Babel, natural and random token prompts. In particular, the Babel prompts are clearly separate from the complete random prompts consisting of random tokens. This suggests that there is a non-trivial structure in the Babel prompts which we further investigate in the next section.

5. The structure of Babel prompts

Perhaps the most surprising aspect of LM Babel is that even though the crafted prompts look like nonsensical set of tokens, LLMs respond to them with predefined coherent text. In Table 1 we include examples of Babel prompts and model responses. In contrast, when prompted with a completely random string of tokens, LLMs would typically display a refusal message saying that the question does not make sense. In this section we aim to understand the nature of this behaviour by examining the characteristics of LM Babel. We analyse patterns in the crafted prompts at the token level and explore if there is a hidden structure in them.

One could suggest that although the optimized inputs appear gibberish, they could still contain trigger words which guide the model into outputting response related to a particular topic. That is, trigger words might be the words from the target texts itself. In Figure 5 we display histograms of the number of tokens from the target string appearing in the constructed prompts across 7B models. On average, Babel prompts contain about 2 tokens from the target string, while the length of the input is 20 tokens. Moreover, there is no statistically significant correlation between the number of target tokens in the prompt and it's success rate (Point-biserial correlation coefficient is 0.06 with p-value 0.09).

In Figure 6, we present the most frequent tokens in Babel prompts for the Vicuna 7B model, with additional data for other models provided in Appendix B.5. Our analysis reveals non-trivial patterns in the context of dataset-specific tokens. For instance, prompts targeting the Wikipedia dataset frequently include tokens like *Wikipedia*, *ipedia*, or *wiki*, which is notable considering these terms never appear in the target strings. Similarly, prompts for the CC-News dataset sometimes incorporate tokens such as *news* or *title*. There is a high likelihood that texts from Wikipedia and news websites were included in the training corpora of these language

Table 5. Average perplexity and entropy of Babel, random, and natural prompts for LLaMA2-7B. Standard deviations for entropy are computed by resampling the corpus containing 70% of prompts.

Metric	Babel	Random	Natural
Avg PPL	11.73 ± 0.60	11.56 ± 0.86	4.10 ± 1.12
Entropy	13.08 ± 0.003	13.35 ± 0.003	12.06 ± 0.012

models. These observations suggest that Babel inputs might be exploiting the models' internal knowledge to establish contextually relevant associations and manipulate model responses effectively.

So far, we found that automatically crafted prompts may contain words related to target text or its source or tokens from the target string itself. Next, we explore whether these seemingly gibberish prompts harbor any underlying structure. In our experiments, we find that perplexity of the Babel inputs is as high as perplexity of random set of the same number of tokens consistent with previous findings (Jain et al., 2023), see Table 5. We further utilize the notion of conditional entropy to analyze the structure of constructed prompts. Conditional entropy, denoted as $H(Y|X)$, measures the average uncertainty in a token Y given the preceding token X:

$$H(Y|X) = -\sum_{x \in X} \sum_{y \in Y} p(x, y) \log \left(\frac{p(x, y)}{p(x)} \right)$$

To compute entropy, we concatenate sampled prompts together into one text corpus. We find that the conditional entropy for Babel prompts is lower than for random strings of tokens, but higher than for natural prompts. This suggests that, while not as structured as natural language, LM Babel does possess a certain degree of order.

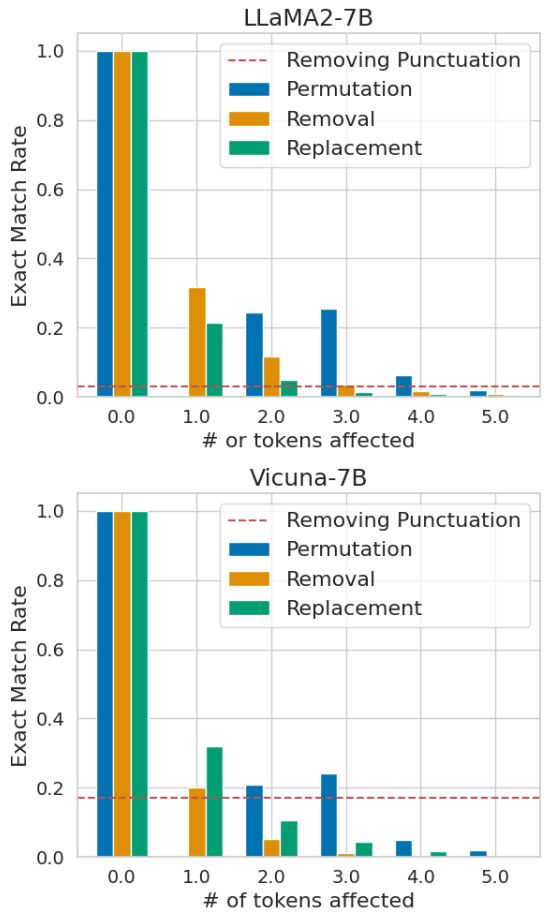

Figure 7. Robustness of successful Babel prompts to token-level perturbations. The plot displays the success rates of prompts subjected to different numbers of removed, replaced, or permuted tokens. For each perturbation method and prompt, we conduct trials with 5 random seeds, averaging the success rates across both seeds and prompts.

6. Robustness of Babel prompts

In this section we examine the robustness of Babel prompts to various token-level perturbations. Our focus is to determine whether the prompt optimization process converges to a flat minimum that leads to the generation of the target string, or if minor token alterations can effectively break the attack.

Our experimental framework encompasses three distinct types of token perturbations: permutation, removal, and replacement. The results of these experiments are illustrated in Figure 7 for the 7B models, and in Appendix Figure 9 for the 13B models. We observe that elimination or substitution of merely a single token results in the failure of over 70% of successful gibberish prompts, and altering two or more tokens neutralizes over 90% of these prompts. While token permutation has a slightly lower impact, it still renders over 95% of inputs ineffective when at least four tokens are permuted.

Given the prevalence of punctuation in the majority of Babel prompts for LLaMA2 models, we explore its role in influencing the adversarial effect. We find that eliminating punctuation elements from the inputs results in the disruption of 97% of gibberish prompts. This finding can be used as another simple defence against out-of-domain adversarial examples along with paraphrasing and retokenization (Jain et al., 2023).

7. Discussion

In this study, we investigate comprehension of adversarial gibberish prompts by large language models. Utilizing gra-

dient guided optimizers, we find that seemingly nonsensical prompts can effectively direct LLMs to generate specific, coherent text. We conduct a systematic analysis of this phenomenon from various perspectives, encompassing the characteristics of the target text, the structural intricacies of Babel prompts, their comparison with natural prompts, and the robustness of these gibberish inputs. We observe that generating longer and more perplex texts poses greater challenges, and that for certain models, Babel prompts attain better loss minima for text generation than their natural counterparts. Moreover, our analysis reveals distinct clustering of Babel prompts in the representation space of the models, highlighting the difference in how models process LM Babel versus natural language inputs. We discern that Babel prompts subtly exploit the models' internal knowledge, leveraging contextually relevant associations, such as using tokens like "wiki" for Wikipedia content generation or "title" for news titles. At the same time we find that Babel prompts tend to be very fragile – minor modifications, such as altering a few tokens or omitting punctuation, can disrupt their effectiveness in up to 97% of cases. Notably, manipulating models into producing harmful content is no more difficult than eliciting benign content, indicating lack of alignment for out-of-distribution prompts. At the same time, fine-tuning models to forget certain content complicates steering the model towards unlearned content. Overall, our work sheds light into non-human languages for interacting with LLMs, which is an important angle for both improving the safety of LLMs and for understanding the inner workings of these models.

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